Southern New Hampshire University

CS 255 Business Requirements Document

System Components and Design

Purpose

What is the purpose of this project? Who is the client and what do they want their system to be able to do?

- DriverPass is a company that wants to help students better prepare for DMV driving tests by offering learning and behind-the-wheel training.
- The purpose is to design a system that allows people to learn, practice, and prepare for their driver's test.
- The system should allow customers and employees to help manage and track appointments, equipment, and learning material such as practice tests and practical exercises.

System Background

What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?

- DriverPass is looking for a single platform where customers can sign up for training, book driving lessons, complete online practice tests, and track how they are progressing.
- The platform will include several user roles, such as customers, the business owner, IT administrators, secretaries, and driving instructors, with each role having access to specific system features.
- All appointments, training packages, customer profiles, instructors, vehicles, and activity logs will be handled in one central system.
- It will connect with the DMV to get updates on driving rules, policies, and practice questions making sure customers always have the latest information available.
- The platform will be cloud based, letting authorized users securely log in through a web browser from various locations, and giving administrators the ability to create reports and manage the system.

Objectives and Goals

What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?

- Allow customers to set up an account, pick a training package, and book, change, or cancel their driving lesson appointments online.
- Offer online practice exams, study materials, and tools for tracking progress so students can get ready for their driving test.

- Let employees handle customer accounts, appointments, instructors, vehicles, and training records using a secure system with different roles.
- Connect with the DMV to keep all materials, laws, and policies up to date, and alert administrators when there are new updates.
- Create reports and keep an activity log that records user actions, appointment updates, and system changes for accountability.
- Let administrators manage user accounts, reset passwords, turn off training packages, and look after the overall security of the system.

Requirements

Nonfunctional Requirements

In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.

Performance Requirements

What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be updated?

- The system needs to be web based and hosted on a secure cloud platform, allowing both customers and employees to access it from computers, tablets, or mobile devices.
- The system is expected to respond promptly when users log-in, schedule or change appointments, take practice tests, check their progress, or create reports.
- It should allow several users to work at the same time without causing noticeable delays or risking data loss.
- When an authorized user makes a change to customer, appointment, payment, or training details, that information should be updated right away.
- Practice tests, DMV rules, policies, and sample questions should all be updated as soon as new information is received from the DMV.
- The system should stay available with very little downtime, while backups, maintenance, and security updates are managed by the cloud service provider.

  Rationale: Performance requirements define how effectively the DriverPass system must operate. Since customers will be scheduling lessons, taking online practice tests, and using training materials through the internet, the system needs to deliver quick response times and handle several users at the same time. Hosting the system in the cloud means users can reach it from various devices, and it also lets updates and maintenance happen with little interruption.

Platform Constraints

What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?

- The DriverPass system needs to be a web based application that works through standard web browsers on Windows, macOS, Android, and IOS devices.

- The system needs to be hosted on a cloud platform, so DriverPass will not have to handle physical servers, backups, or most of the system's maintenance.
- The back end will use a centralized database that stores customer information, employee accounts, appointments, training packages, test results, driver notes, vehicle details, and records of activity.
- A secure internet connection will be necessary for the system to access or update current information and to communicate with the DMV system.
- The system will let authorized users download reports and selected data in file formats that can be opened in programs like Microsoft Excel.

  Rationale: Platform constraints identify the operating environments and the technology that supports the system. A browser based, cloud hosted system lets customers and employees use DriverPass from various computers or mobile devices, and they do not have to install any special software. A central database is needed to keep records consistent and avoid duplicate or conflicting information.

Accuracy and Precision

How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?

- The system needs to identify users by their unique usernames or email addresses, assigning permissions according to their roles such as customer, secretary, driving instructor, owner, or IT administrator.
- Passwords must be case sensitive. At the same time, names, addresses, and email addresses should be managed in a consistent way to avoid creating duplicate customer records.
- The system must check that all required information is provided before it accepts any registration, payment, appointment, or account update.
- It should also confirm that lesson times, instructors, and vehicles are available before an appointment is finalized.
- The system is expected to keep an accurate record of who created, changed, or canceled an appointment, including the date and time when the action took place.
- Whenever there are repeated login failures, system errors, failed DMV updates, scheduling conflicts, or unusual activity on an account, the system should notify the IT administrator.

  Rationale: Role based identification stops users from accessing features outside their responsibilities, while validation checks help reduce errors in customer, appointment, and payment information. Administrative alerts let issues be fixed before they impact customers or business operations.

Adaptability

Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?

- The system should let the IT administrator add, remove, disable, or change user accounts and assigned roles without needing to modify the system's code.
- The IT administrator needs full access to manage accounts, reset passwords, block former employees, check activity logs, and troubleshoot system problems.
- The system should let authorized users turn training packages on or off without needing any software changes.
- The cloud based system must stay compatible with updates to common web browsers, mobile devices, and operating systems.
- The system should be built so DMV rules, practice questions, and training materials can be updated without changing the main system structure.
- Major features, like adding completely ne package modules, might still need help from a developer or systems analyst.

  Rationale: Adaptability requirements make sure DriverPass can handle routine business changes without needing a complete rebuild. User accounts, permissions, packages, and training content should be manageable with administrative tools, and the system should keep working as browsers and operating systems change. For more complicated changes, technical support may be needed, since Liam agreed that adding or removing major modules could be included in a future release.

Security

What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a "brute force" hacking attempt? What happens if the user forgets their password?

- The system must require each user sign in using a unique username or email address, along with a secure, case sensitive password.
- Role based access should be implemented so that customers, secretaries, instructors, management, and the IT administrator can only reach the information and features necessary for their roles. All data exchanged between users and the cloud server needs to be encrypted over a secure connection, and special attention to login, customer, and payment details.
- Password and credit card data must be stored securely and never shown in plain text.
- If there are several failed login attempts, the system should temporarily lock the account and alert the IT administrator about possible unauthorized access.
- Customers and employees who forgot their passwords must have a way to request a secure password reset link or go through a verification process.
- The IT administrator should have the ability to reset passwords, disable accounts, and immediately block access for employees who no longer work at the company.
- The system must keep an activity log that records who accessed, created, changed, or canceled records

  Rationale: Security matters because DriverPass will keep personal, payment, account, and scheduling information. Secure logins, encrypted connections, access controls, and account lockouts help block unauthorized access. Password recovery and administrative controls also let

legitimate users regain access, while giving DriverPass the ability to react quickly to suspicious activity or changes in staff.

Functional Requirements

Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with "The system shall . . ." For example, one functional requirement might be, "The system shall validate user credentials when logging in."

- The system shall let customers create and manage their own accounts.
- The system shall check user credentials during login.
- The system shall let users reset their password if they forget it
- The system shall let customers choose and buy a DriverPass training package.
- The system shall let customers and authorized employees schedule, change, or cancel driving lesson appointments.
- The system shall assign instructors and vehicles that are available to each scheduled driving lesson.
- The system shall avoid scheduling conflicts by checking if the instructor, vehicle, and appointment time are free.
- The system shall give customers access to online practice tests and learning materials that match the package they chose.
- The system shall keep track of and show customer progress, including test scores, completion status, and lesson history.
- The system shall let the IT administrator manage user accounts, reset passwords, disable accounts, and set user roles.
- The system shall make reports that authorized users can view, print, or download.

Rationale: Functional requirements describe what the DriverPass system needs to do to serve both customers and employees. These actions include online registration, scheduling appointments, drivers training, practice testing, managing users, reporting, and system administration.

User Interface

What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?

- The system should offer a web based interface that users can reach from computers, tablets, or mobile devices by using a web browser.
- Customers need to be able to set up accounts, sign in, handle appointments, take practice tests, check their progress, and look over instructor feedback.
- Employees such as the secretary, instructors, managers, and IT administrator should have interfaces tailored to their roles. These should let them handle their specific tasks, like booking lessons, keeping track of customer records, adding driver notes, preparing reports, and managing user accounts.

- The interface should include straightforward forms for registration, scheduling, payments, and contact details, as well as clear navigation and progress indicators for both practice tests and driving lessons.

  Rationale: The user interface must be straightforward and easy to use, so customers and employees can finish their tasks without trouble. A web based interface makes it possible for users to get into the system from different devices, while role based screens make sure each person only sees the features they need for their own work.

Assumptions

What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?

- Customers and employees will need to have a stable internet connection and a computer, tablet, or mobile device that uses a supported web browser.
- Users must enter correct customer, payment, pickup, and scheduling details when setting up accounts or making appointments.
- DriverPass will keep enough instructors and vehicles available to meet the demand for scheduled driving lessons.
- The DMV will use a method that works with the system to send updated rules, policies, and practice test details.
- The cloud service provider will handle system hosting, backups, security updates, and general server maintenance.

  Rationale: These assumptions describe conditions that the system relies on but that were not completely detailed in the interview. The DriverPass system needs internet access, supported devices, correct user data, available training resources, and dependable cloud and DMV services to work as expected.

Limitations

Any system you build will naturally have limitations. What limitations do you see in your system design? What limitations do you have as far as resources, time, budget, or technology?

- The system will need an internet connection to get up to date information, book appointments, and receive DMV updates.
- The system will rely on the cloud service and DMV system being available, if either one is down, some features might not work.
- Customers who do not have internet access or a supported web browser will have to contact the DriverPass office to sign up or make appointments.
- The first version of the system will let administrators turn training packages on or off, but making brand new packages or major features will need help from a developer or systems analyst.
- The system's speed and reliability could be affected during planned maintenance or if there are unexpected service outages.

Rationale: These limitations show constraints that cannot be removed through system design. Since DriverPass depends on cloud hosting and DMV integration, some features rely on outside services being up and running. Also, the first project scope does not let nontechnical users create new system modules or major features without support from a developer.

Gantt Chart

